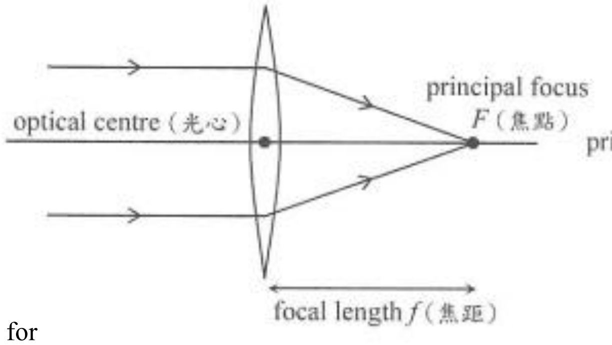
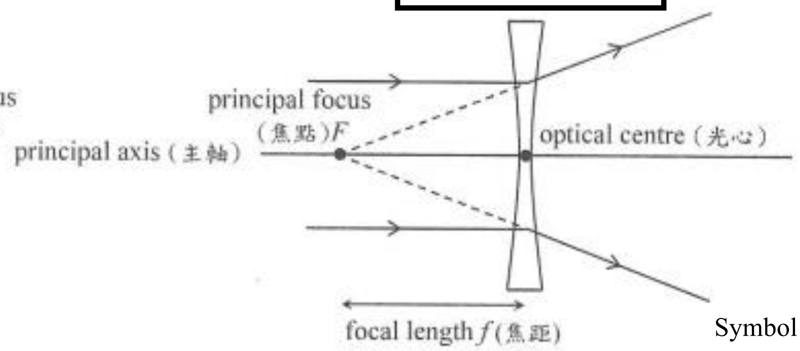
薈進教育中心
Unit Education
Tel: 98583618

Convex lens



Concave lens



Symbol for convex lens

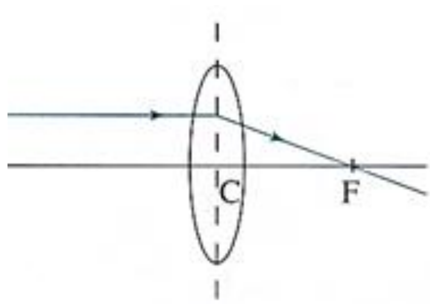
As incident rays parallel to principal axis, refracted rays **converge** to focus.
(Convex lens = converging lens)

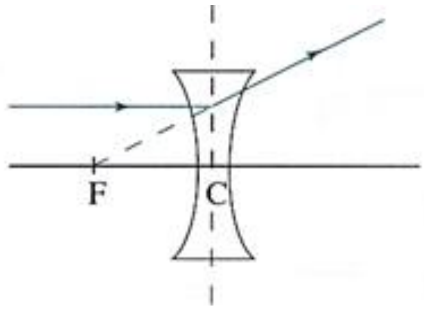
As incident rays parallel to principal axis, refracted rays **diverge** to focus.
(Concave lens = diverging lens)

Symbol for concave lens

Rule 1: A ray parallel to the principal axis is bent to pass through F on the other side of lens

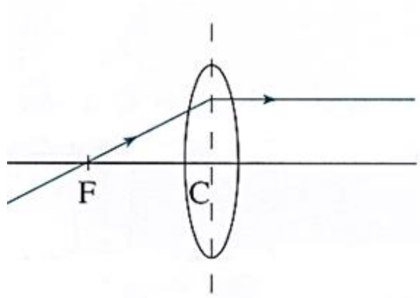
平衡入 , Focus(F) 出

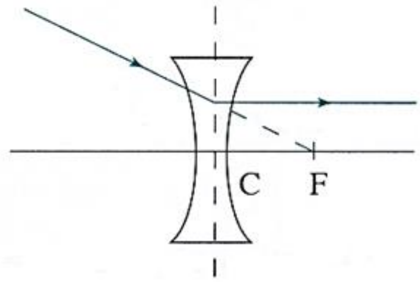




Rule 2: A ray passing through F emerges parallel to the principal axis

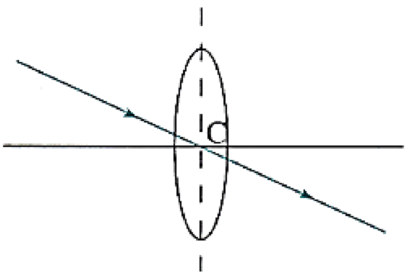
Focus(F) 入 , 平衡出

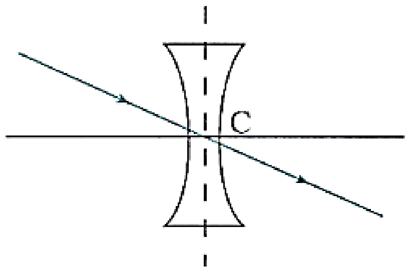




Rule 3: A ray passing through C travels straight on

C 入 C 出 , 直入直出

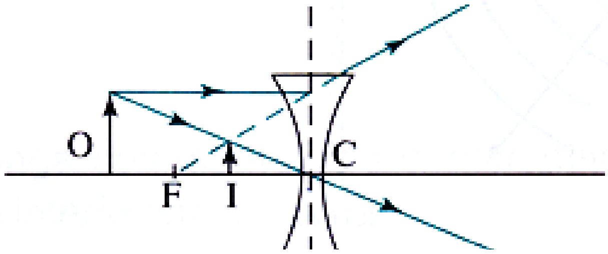


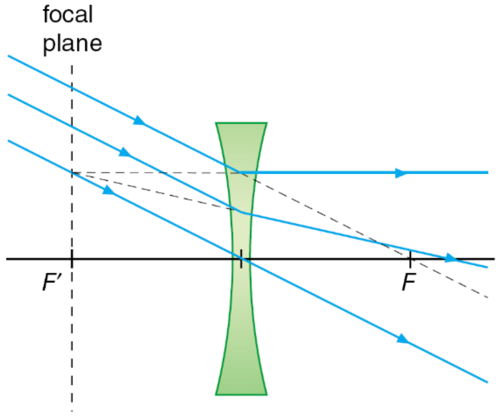


Concave lens

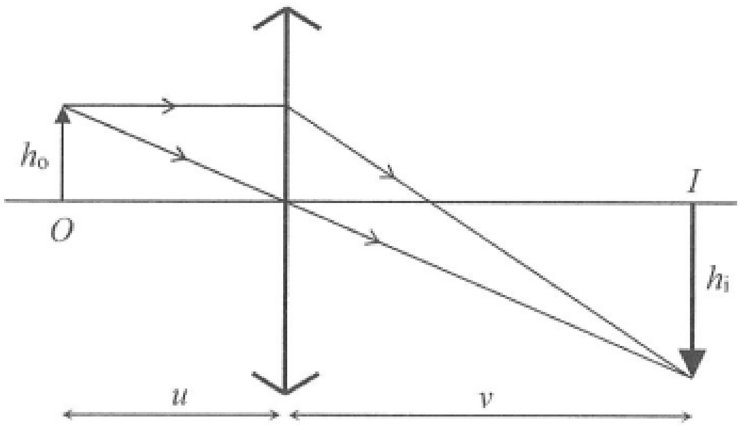
Nature of image:

- Virtual
- Erect
- Diminished
  (form between C and F)
- **Remarks:**
  All incident parallel rays after refraction by concave lens must diverge from a focus on the focal plane.
- **Application** of concave lens:
  (i) Peep-hole lens
  (ii) Spectacles for correcting short-sighted eye





Magnification



Linear magnification (M)

$$M = \frac{v}{u} = \frac{h_i}{h_o}$$

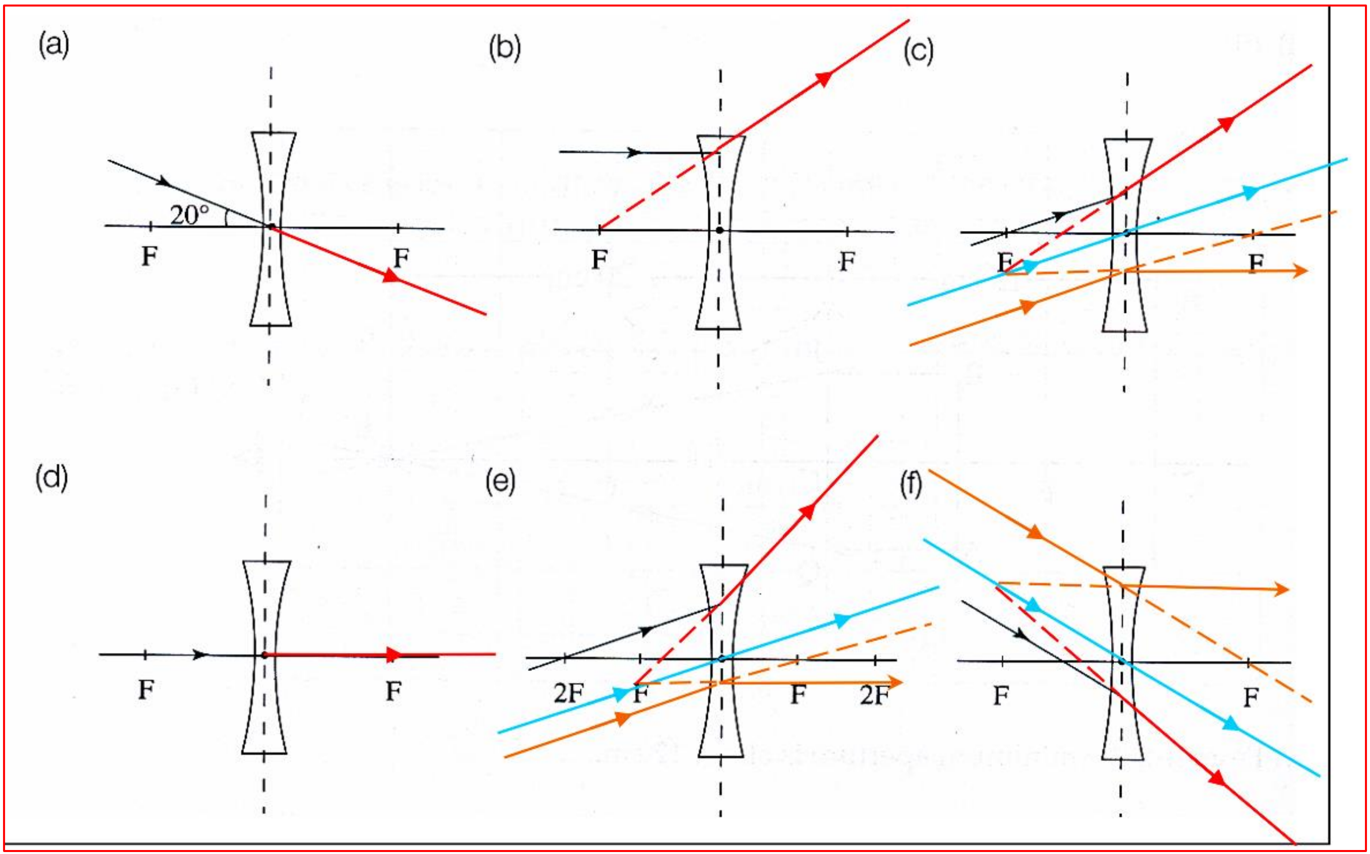
v : image distance
u : object distance
h_i : image height
h_o : object height

Lens formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

v : image distance (**real image: +ve, virtual image: -ve**)
u : object distance (**always +ve**)
f : focal length (**convex lens: +ve, concave lens: -ve**)

1. Draw the refracted rays.



2. Which of the following is correct about the images formed by concave lenses?

   A The images must be real, erect and magnified.

   B The images must be real, erect and diminished.

   C The images must be virtual, inverted and magnified.

   D The images must be virtual, erect and diminished.

The image of concave lens must be virtual, erect and diminished.

D

3. Which of the following statements correctly describes a concave lens?

   (1) The image of a real object formed by the lens may be real or virtual depending on the distance.

   (2) The image of an object formed by the lens may be magnified or diminished depending on the object distance.

   (3) It is used for the correction of short-sightednes

   A. (1) only B. (3) only

   C. (1) and (2) only D. (2) and (3) only

(1): image formed by a concave lens must be virtual.

(2): image formed by a concave lens must be diminished.

B

4. Which of the following statements about the image formed by a concave lens is/are correct?

   (1) It is between the lens and object.

   (2) It is always diminished.

   (3) It will be formed at infinity if the object is located at the focus of the lens.

   A (2) only

   B (1) and (2) only

   C (1) and (3) only

   D (2) and (3) only

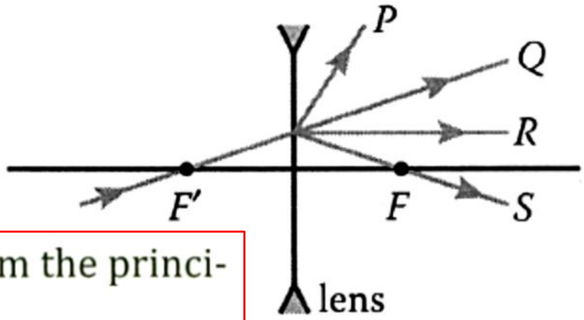
警

(3): The image will still be erect and diminished, the exact position can be found with a aid of two light rays, one parallel to the horizontal, prior to refraction, and one penetrating the center of the focus.

B

5. F and F' are the foci of the lens shown. Which of the following best shows the path of the emergent ray?

A. P
B. Q
C. R
D. S

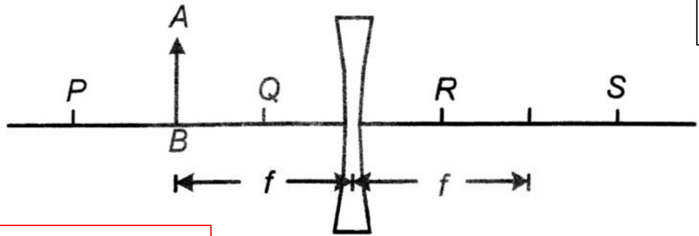


A

A The ray should deflect away from the principal axis for a concave lens.

6. An object AB is placed at a distance of one focal length f in front of a concave lens as shown in the figure. What are the position and the nature of the image?

	Position	Nature
A.	at P	virtual and erect
B.	at Q	virtual and erect
C.	at R	real and inverted
D.	at S	real and inverted



B

The image must be virtual and erect with a concave lens. Its position also must be between the object and the lens

7. A lens is placed in front of a book as shown. Which of the following statements are correct?
(1) The lens is a concave lens.
(2) The image is virtual.
(3) The image distance is smaller than the focal length of the lens.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

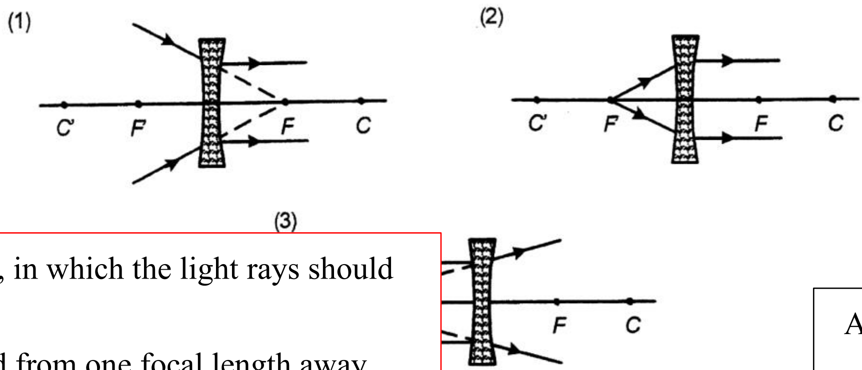
D

D (1) and (2) are correct. As the image is erect, it must be virtual. And only a concave lens forms a diminished virtual image.

(3) is also correct. An image formed by a concave lens must lie between F and C. It is on the focal plane when the object is at infinity.

8. If F, F' are foci and C, C' are both at a distance of two times the focal length from the lens, which of the following ray diagrams is/are correct?

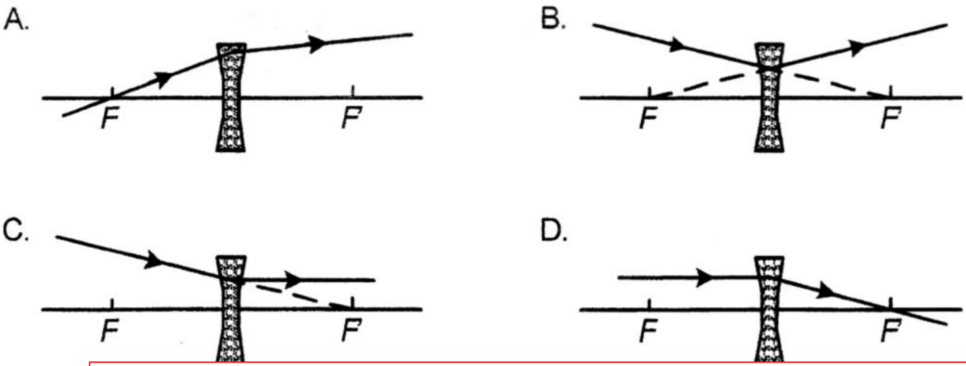
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only



A

(2): the concave lens is a diverging lens, in which the light rays should bend toward outside.
(3): The dotted line should be originated from one focal length away from the lens, i.e. at F.

9. If points F and F’ represent the focal length of a concave lens, which of the following ray diagrams correctly shows the path of a light ray through the lens?

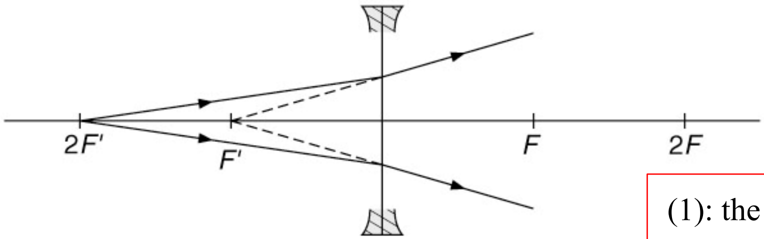


C

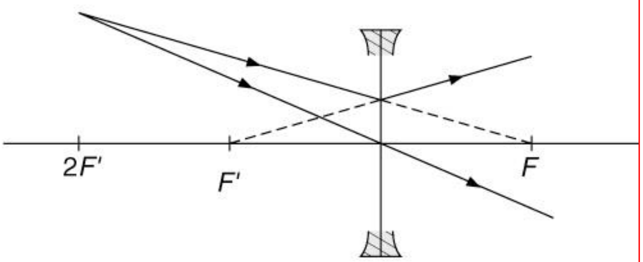
A and D are demonstrating converging light rays, in which require convex lens to show such pathways.
B is demonstrating the effect of a concave lens, but the refracted light ray should be parallel to the horizontal since the initial light ray is pointed towards the focus on the other side of the lens.

10. If F and F ′ are the foci of a concave lens, which of the following ray diagrams is/are **incorrect**?

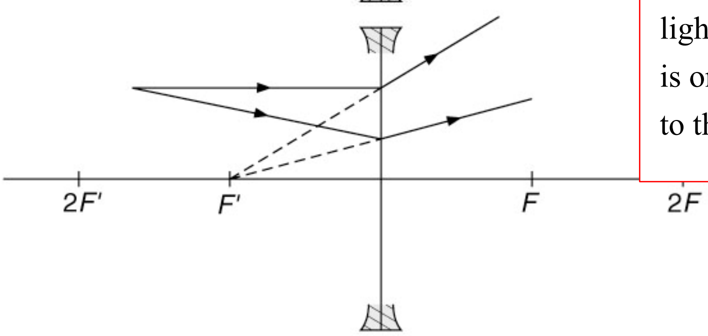
(1)



(2)



(3)

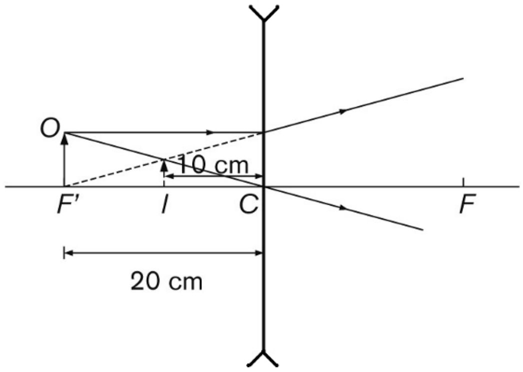


(1): the direction of the refracted light ray should not be originated from the focus since the initial light ray is not parallel to the horizontal.
(2): The superior light ray should be parallel to the horizontal after refracted, since the original light ray projects to the focus on the other side of the lens.
(3): The inferior light ray should have a refracted light ray with a smaller elevation since such pathway is only correct when the original light ray is parallel to the horizontal.

A (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

D

11. According to the figure, which of the following about the image I is correct?



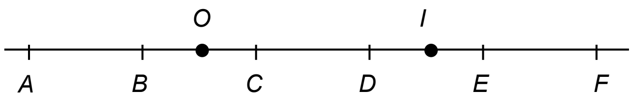
	Magnification	Image nature
A.	0.5	real
B.	0.5	virtual
C.	2	real
D.	2	virtual

$$m=\frac{v}{u}=\frac{10}{20}=0.5$$

The image must be virtual, erect, diminished.

B

12. In the following diagram, intervals *AB*, *BC*, *CD*, *DE* and *EF* are of equal length. An object is placed at *O*. When a lens is put in front of the object, an image of the object is formed at *I*.



B

Which of the following show(s) the possible position(s), type(s) and focal length(s) of the lens?

	Position	Type	Focal length
(1)	C	Convex lens	about 1 interval
(2)	F	Concave lens	about 2 interval
(3)	D	Convex lens	about 1 interval

A. (1) only

B. (2) only

C. (1) and (3) only

D. (2) and (3) only

For option (1), the object distance is smaller than the focal length of the convex lens, so the image should be formed on the same side as the object.

∴ (1) is incorrect.

For option (2), a concave lens should form an image within a focal length from the lens. The image should be on the same side as the object.

∴ (2) is correct.

For option (3), the object distance is between *f* and 2*f*, so the image distance should be greater than 2*f*. The image and the object should be on opposite sides of the lens.

∴ (3) is incorrect.

13. An object placed in front of a converging lens of focal length 10 cm forms an image 30 cm from the lens. If the image is real, what is the distance of the object from the lens?

A. 7.5 cm
B. 8.5 cm
C. 12.5 cm
D. 15 cm

D $\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$

$\Rightarrow \frac{1}{u}+\frac{1}{+30}=\frac{1}{+10}$ (Image distance is positive as the image is real.)

$\Rightarrow u=+15$ cm

D

14. An object placed in front of a converging lens of focal length 10 cm forms an image 30 cm from the lens. If the image is virtual, what is the distance of the object from the lens?

A. 7.5 cm
B. 8.5 cm
C. 12.5 cm
D. 15 cm

A

A $\frac{1}{u}+\frac{1}{-30}=\frac{1}{+10}$ (Image distance is negative as the image is virtual.)

$\Rightarrow u=+7.5$ cm

15. An object is placed at the focus of a concave lens of focal length 10 cm. What is the magnification of the image formed ?

A. 0.5
B. 1.0
C. 2.0
D. infinite

By $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$ $\quad\therefore \frac{1}{(-10)}=\frac{1}{(10)}+\frac{1}{v}$ $\quad\therefore v=-5$ cm

Linear magnification : $m=\frac{v}{u}=\frac{(5)}{(10)}=0.5$

A

16. A man is reading some print with a concave lens. The lens is 9 cm in front of the print and the linear magnification of the image is 0.4. What is the distance between the print and its image?

A. 3.6 cm
B. 5.4 cm
C. 9 cm
D. 12.6 cm

B

A man is reading...

B u = 9 cm and $v = 0.4u$ = 3.6 cm. Since the object and the image are on the same side of the lens, their separation = 9 – 3.6 = 5.4 cm.

17. An object placed 15 cm from a lens produces a virtual image which is magnified 2 times. The lens is a

A. concave lens of focal length 10 cm.
B. convex lens of focal length 10 cm.
C. concave lens of focal length 30 cm.
D. convex lens of focal length 30 cm.

D

By $v=mu=(2)(15)=30$ cm

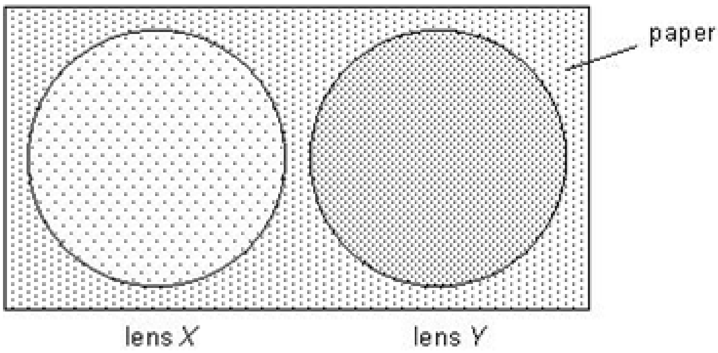
By $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$ $\quad\therefore \frac{1}{f}=\frac{1}{(15)}+\frac{1}{(-30)}$ $\quad\therefore f=+30$ cm

The lens is convex with focal length 30 cm.

18. Dots are printed on a piece of paper evenly. Sophia holds two lenses *X* and *Y*, both of very long focal lengths, above the paper. The following figure shows what she sees.

(a) Determine whether lenses *X* and *Y* are concave or convex.

(b) For each of the lenses, arrange the lens, the paper and the image formed by the lens in ascending order of their distances from Sophia.



(a) *X* is a convex lens.
Y is a concave lens.
(b) For *X*: lens, paper, image
For *Y*: lens, image, paper

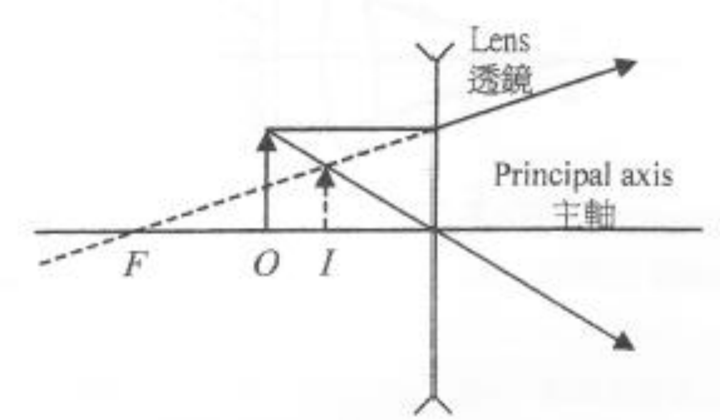
19. A student holds a lens above a picture and the image observed is shown in the figure.

(a) What kind of lens is used by the student? Explain your answer.

(b) Sketch a ray diagram to show how the image in the figure is formed.

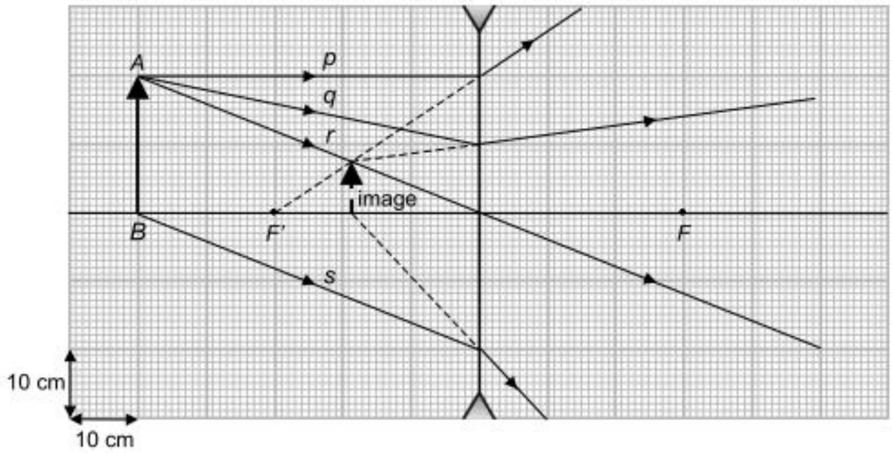
(a) The lens is a <u>diverging</u> (<u>OR</u> <u>concave</u>) lens because

the image formed is erect and diminished.



20. An object AB is placed in front of a concave lens as shown.

4(a) 3A



(a) Draw the refracted rays of p, q, r and s and the image of AB. (3 marks)

(b) State two natures of the image (2 marks)

(c) If the object AB is now moved to F', will the image move towards the lens or away from the lens ? (1 mark)

b) The image is virtual / erect / diminished. (any two) 2A

c) Towards the lens 1A

21. An object AB is placed in front of a lens as shown.

(a) What kind of lens is used ? (1 mark)

(b) Can the image formed be captured by a screen ? (1 mark)

(c) (i) Complete the ray diagram to locate the image of AB. (2 marks)

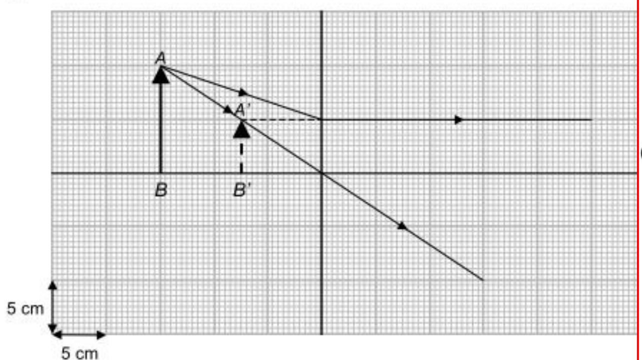
(ii) What is the magnification of the image? (2 marks)

(d) Without using lens formula, find the focal length of the lens. (2 marks)

(a) Concave lens 1A

(b) Virtual 1A

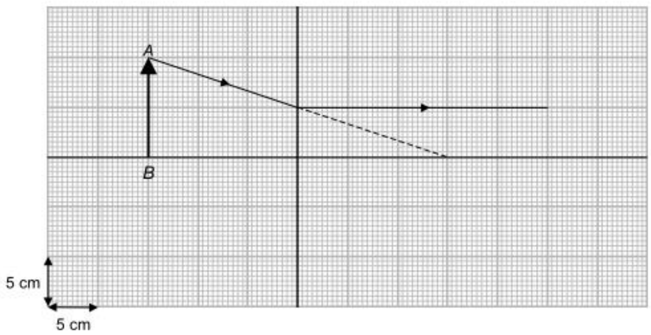
(c) (i)



(ii) Magnification $= \frac{v}{u} = \frac{5}{10}$ 1M

$= 0.5$ 1A

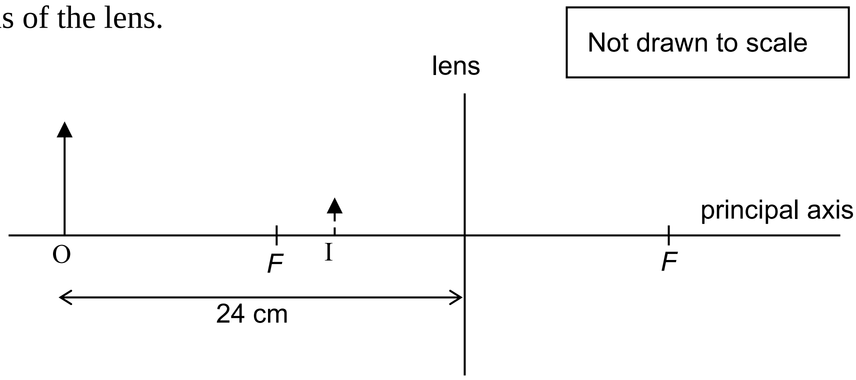
(d)



The focal length of the lens $= 3 \times 5$ 1M

$= 15$ cm 1A

22. In the following figure, the positions of an object O and its images I formed by a lens are shown. F is the focus of the lens.



The object is 9 cm tall and its image is 3 cm tall.

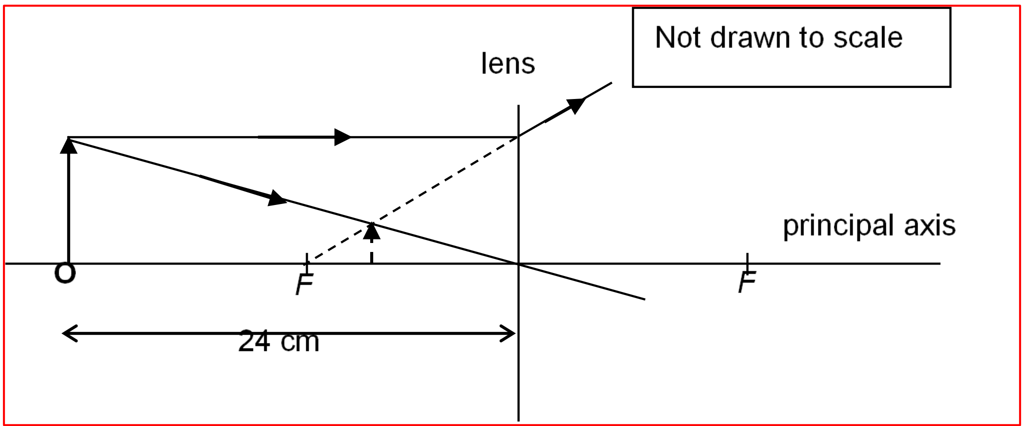
(a) Find the magnification of the image.

(b) (i) Find the image distance.

(ii) Determine the focal length and the type of the lens.

(c) Draw two rays in the figure to show how the image is formed by the lens.

(d) Explain whether the image can be captured by a screen.



(a) Magnification $= \dfrac{h_i}{h_o} = \dfrac{3}{9} = \dfrac{1}{3}$

(b) (i) By $m = \dfrac{v}{u}$,

image distance $= mu = \dfrac{1}{3} \times 24 = 8$ cm

(ii) By the lens formula,

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

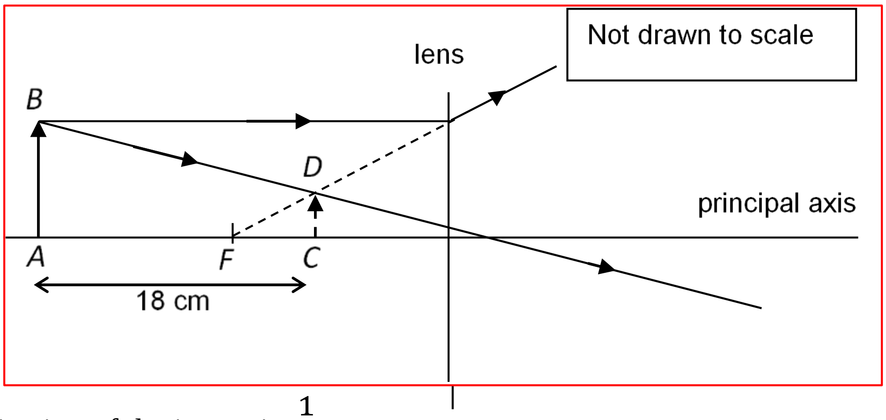
$$\frac{1}{24} + \frac{1}{-8} = \frac{1}{f}$$

$f = -12$

The focal length of the lens is 12 cm and it is a concave lens.

(d) The image cannot be captured by a screen since it is virtual.

23. An object AB and its image CD formed by a lens are 18 cm apart as shown in the figure below.



The magnification of the image is $\frac{1}{4}$.

(a) Find the object and the image distances.

(b) Determine the focal length and the type of the lens.

(c) In the figure above,

(i) draw two light rays to show how the image CD is formed,

(ii) indicate the position of the focus of the lens with a letter F.

(d) A student claims that if the object AB is placed at the focus of the lens, the image I will be formed at infinity. Give your comment.

(a) By $m = \frac{v}{u}$,

$v = mu = \frac{1}{4}u$ ....(1)

It is given that:

$u - v = 18$ ....(2)

Substitute (1) into (2):

$u - \frac{1}{4}u = 18$

$u = 24$ cm

Substitute $u = 24$ into (1):

$v = \frac{1}{4}u = \frac{1}{4} \times 24 = 6$ cm

The object distance is 24 cm and the image distance is 6 cm.

(b) Apply the lens formula,

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{-6} + \frac{1}{24} = \frac{1}{f}$$

$f = -8$ ∴ The focal length of the lens is 8 cm. It is a concave lens.

(d) The student is wrong.

Since the lens is concave,

the position of the images is always between the lens and the objects.

24. A lens of focal length 30 cm is placed 20 cm in front of a toy. The image of the toy formed by the lens is shown below.

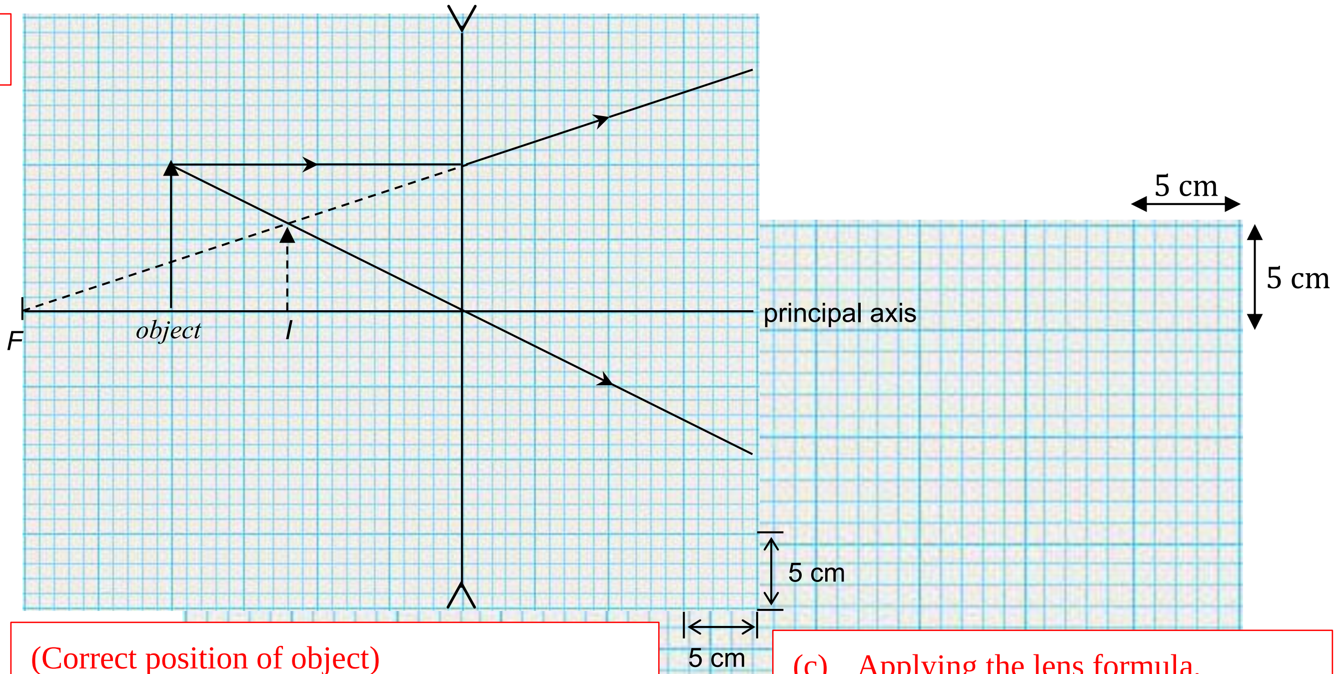
(a) Which type of lens is used? (1 mark)

(b) Draw a ray diagram on graph paper to show how the image of the toy is formed by the lens.(4 marks)

(c) The lens is then moved 10 cm away from the toy. Use the lens formula to find the image distance. (2 marks)

(a) Concave lens

(b)



(Correct position of object)
(Correct application of construction rules)
(Correct position of the image (12 cm ± 1 cm))

(c) Applying the lens formula,

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$$

$$\frac{1}{30}+\frac{1}{v}=\frac{1}{-30}$$

$$v=-15$$

∴ The image distance is 15 cm.